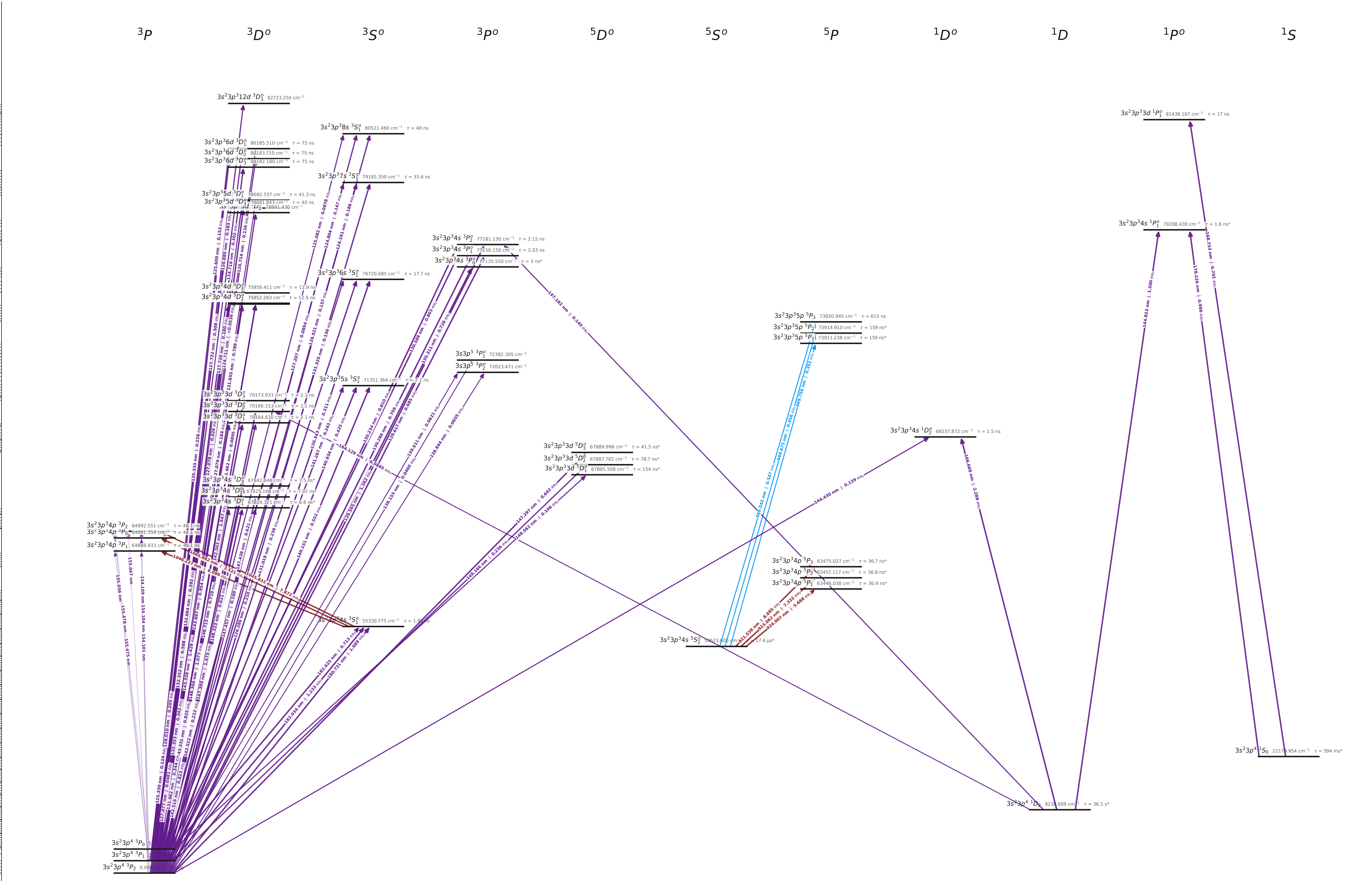


Sulfur

S (S I). Strongest classified transitions below 2 μm (NIST): 47 levels, 81 lines.

S⁺ ionisation limit 83559.17 cm⁻¹ = 10.36002 eV (119.676 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ P ₂ ^o	129.5653	–	2313.8321	1.362	3.457e+08	74.16	74.2	129.6 nm ground-state lines of the 1299 A multiplet used...
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ P ₁ ^o	129.6174	–	2312.9033	0.685	1.456e+08	78.25	29.6	129.6 nm ground-state lines of the 1299 A multiplet used...
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4p ³ P ₂	154.1009	–	1945.429689(27)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4p ³ P ₀	154.1037	–	1945.393840(27)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4p ³ P ₁	154.1095	–	1945.321233(27)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4p ³ P ₂	155.0472	–	1933.55618(7)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₁ – 3s ² 3p ³ 4p ³ P ₁	155.0559	–	1933.447755(27)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₀ – 3s ² 3p ³ 4p ³ P ₂	155.4752	–	1928.233794(27)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₀ – 3s ² 3p ³ 4p ³ P ₀	155.4781	–	1928.197819(27)	–	–	3.452	–	Doppler-free two-photon excitation
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4s ³ S ₁ ^o	180.7312	–	1658.7749	1.689	3.264e+08	84.88	61.2	180.731 nm
3s ² 3p ³ 4s ³ S ₁ ^o – 3s ² 3p ³ 4p ³ P ₂	1045.8308	1045.5443	286.6548	7.872	2.195e+07	3.452	100	1045.5-1045.9 nm near-infrared triplet
3s ² 3p ³ 4s ³ S ₁ ^o – 3s ² 3p ³ 4p ³ P ₀	1045.9618	1045.6753	286.6189	3.521	2.195e+07	3.452	100	1045.5-1045.9 nm near-infrared triplet
3s ² 3p ³ 4s ³ S ₁ ^o – 3s ² 3p ³ 4p ³ P ₁	1046.2267	1045.9401	286.5464	6.099	2.194e+07	3.452	100	1045.5-1045.9 nm near-infrared triplet

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

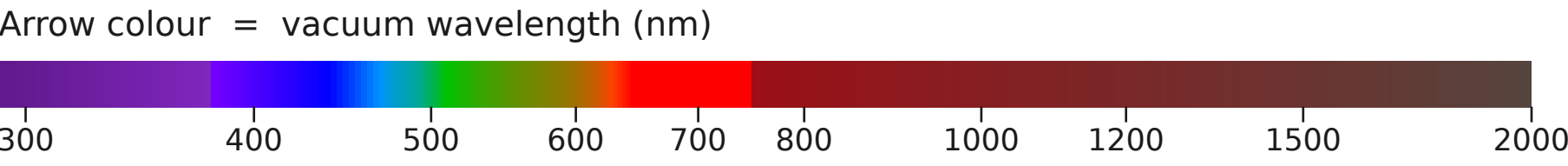
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



³³S hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3s ² 3p ⁴ ³ P ₂	130.090	-31.600	F=1/2: -613.05 F=3/2: -390.27 F=5/2: -45.30 F=7/2: +382.37
3s ² 3p ⁴ ³ P ₁	-4.620	15.800	F=1/2: +31.30 F=3/2: -11.18 F=5/2: -2.98

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3s ² 3p ⁴ ³ P ₂ – 3s ² 3p ³ 4p ³ P ₂	154.101	34-32	537(12)

Shift = ν(first isotope) – ν(second isotope).